

Amazon S3 IAM Policy for Bucket Access

To Begin with the Lab

Summary of the Lab

In this lab, **IAM Policies** were used to control access to Amazon S3 buckets without using Bucket Policies. Three separate S3 buckets were created to demonstrate **Read-Only**, **Write-Only**, and **Full Access** permissions. A new IAM user was created with no initial permissions, and custom IAM policies were attached one by one to observe how each permission affected the user's ability to list buckets, read objects, upload files, and delete objects. The lab also demonstrated the importance of the **Amazon Resource Name (ARN)** while writing IAM policies and highlighted the difference between IAM Policies and S3 Bucket Policies.

Understanding IAM Policies for Amazon S3

An **IAM Policy** is an identity-based policy that grants permissions to IAM users, groups, or roles. When attached to an IAM identity, it determines what actions the identity can perform on AWS resources. For Amazon S3, IAM Policies commonly specify:

- **Effect** – Allow or Deny
- **Action** – S3 operations such as GetObject, PutObject, or DeleteObject
- **Resource** – The ARN of the bucket or object

Unlike **Bucket Policies**, which are attached to buckets, **IAM Policies** are attached to users or roles. In this lab, access was provided entirely through IAM Policies while keeping Bucket Policies empty.

Example:

An IAM user named **S3User** receives:

- Read-only access to **Bucket A**
- Write-only access to **Bucket B**
- Full access to **Bucket C**

The same user experiences different permissions depending on which bucket is accessed because each bucket has a different IAM policy attached through the user.

Lab Steps

Step 1: Create S3 buckets

Create three S3 buckets:

- Read-Only Bucket

Amazon S3 > Buckets > Create bucket

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

cloudfreeks-read-only

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is controlled by bucket policies and IAM permissions.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects is controlled by bucket policies and IAM permissions.

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- ## Write-Only Bucket

Amazon S3 > Buckets > Create bucket

Create bucket [Info](#)
Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
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Bucket name [Info](#)

cloudfreeks-write-only

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

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- ## Full Access Bucket

- Upload a sample file (sample_file.txt) into each bucket for testing.

- Open **IAM** → **Users**
- Create a new user (for example, S3User)

- Enable AWS Console access
- Set a custom password

User name
SSUser

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, =, @, _ (hyphen)

☒ **Provide user access to the AWS Management Console - optional**
In addition to console access, users with `signing:console:development` permissions can use the same console credentials for programmatic access without the need for access keys.

Are you providing console access to a person?

User type

☐ Specify a user in Identity Center - Recommended
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.

☒ **I want to create an IAM user**
IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ **Custom password**
Enter a custom password for the user.

Must be at least 8 characters long
Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + = (hyphen) = [] { } ' "

☐ Show password

☐ Users must create a new password at next sign-in - Recommended
Users automatically get the `IAMUserChangePassword` policy to allow them to change their own password.

If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user.

- Do not attach any permissions initially

Review and create
Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name
SSUser

Console password type
Custom password

Require password reset
No

Permissions summary

Name	Type	Used as
No resources		

Tags - optional
Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.
No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create user](#)

- Sign in using the IAM user credentials

ap-southeast-2.signin.aws.amazon.com/oauth?client_id=arn%3Aaws%3Asignin%3A%3A%3Aconsole%2Fcanvas&code_challenge=...

Provide feedback Multi-session disabled English

aws

IAM user sign in

Account ID or alias (Don't have?)

☐ Remember this account

IAM username
S3User

Password

☐ Show Password [Having trouble?](#)

Sign in

[Sign in using root user email](#)

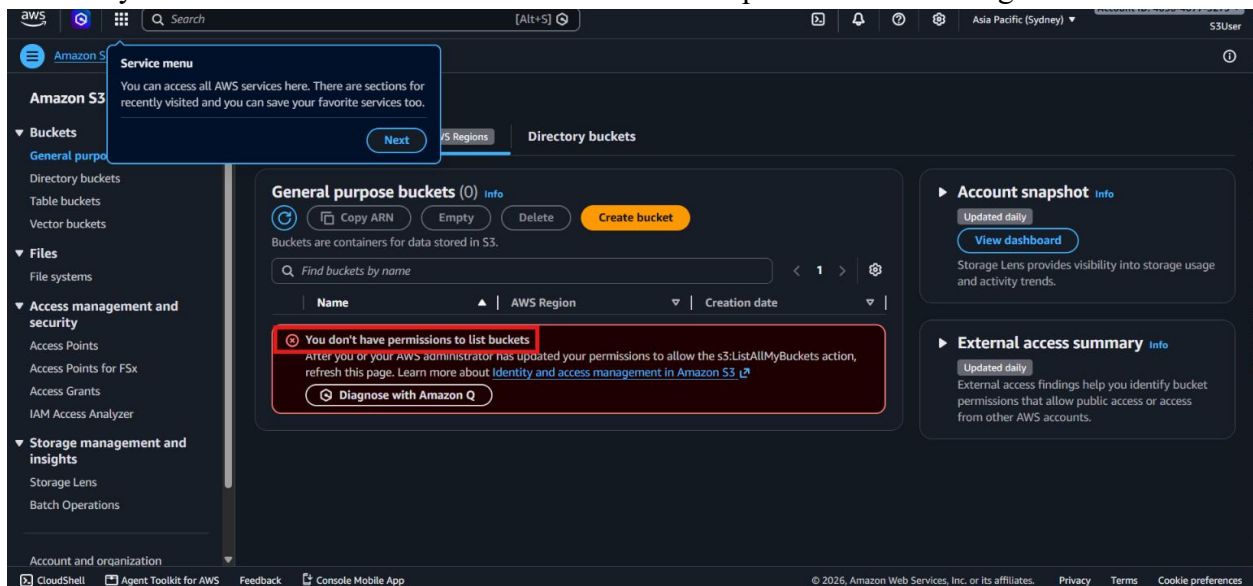
[Create a new AWS account](#)

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AI for ISVs on AWS
Build agents into your SaaS, shift pricing models, and monetize AI on AWS Marketplace

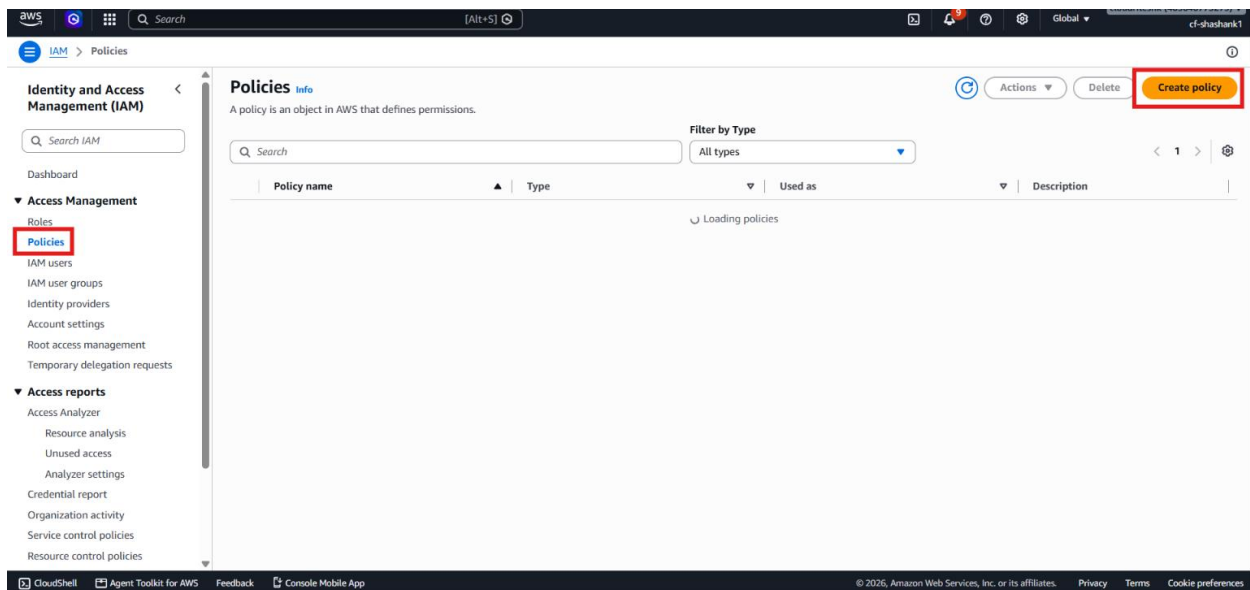
Explore AI for ISVs →

- Verify that the new user cannot access S3 because no permissions are assigned.

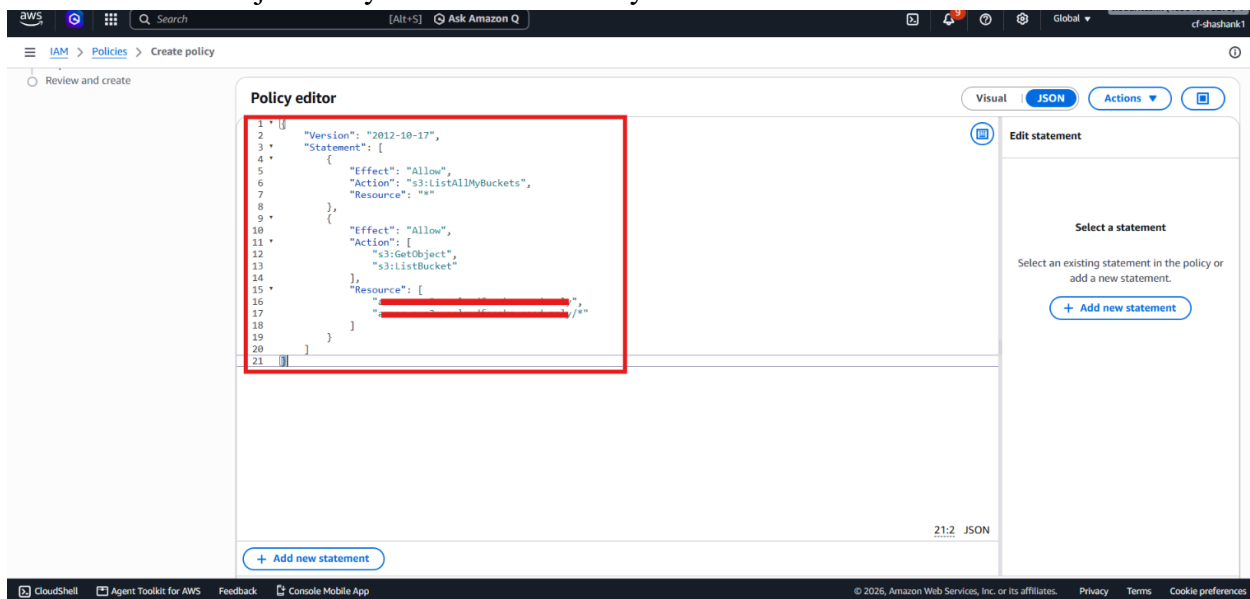


Step 3: Create a Read-Only IAM Policy

- Open IAM → Policies
- Create a new **Customer Managed Policy**



- Use the JSON editor
- Configure permissions to:
 - List all S3 buckets
 - List objects
 - Read objects only from the Read-Only bucket



- Create the policy and set the required name for policy.

Read-only-s3-policy Info

Policy details

Type	Creation time	Edited time	ARN
Customer managed	July 20, 2026, 13:37 (UTC+05:30)	July 20, 2026, 13:37 (UTC+05:30)	arn:aws:iam::123456789012:policy/Read-only-s3-policy

Permissions defined in this policy Info

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Allow (1 of 472 services)

Service	Access level	Resource	Request condition
s3	Limited: List, Read	Multiple	None

- Attach the policy to the IAM user by opening user and clicking on **Add Permissions**.

S3User Info

Summary

ARN	Console access	Access key 1
arn:aws:iam::123456789012:user/S3User	Enabled without MFA	Create access key

Created: July 20, 2026, 13:15 (UTC+05:30)

Last console sign-in: Today

Permissions policies (0)

Permissions are defined by policies attached to the user directly or through groups.

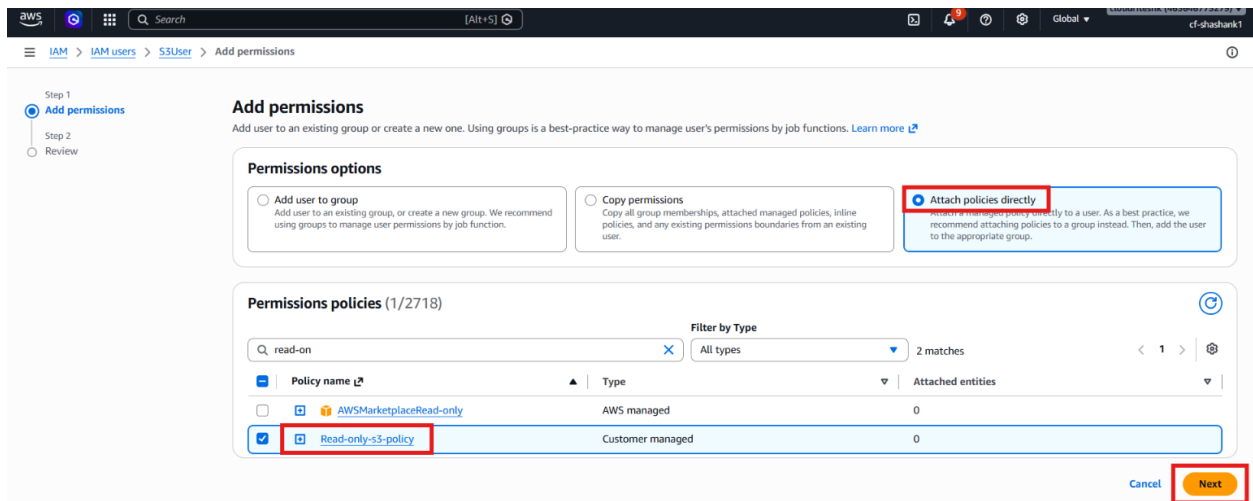
Filter by Type: All types

Policy name	Type	Attached via
No resources to display		

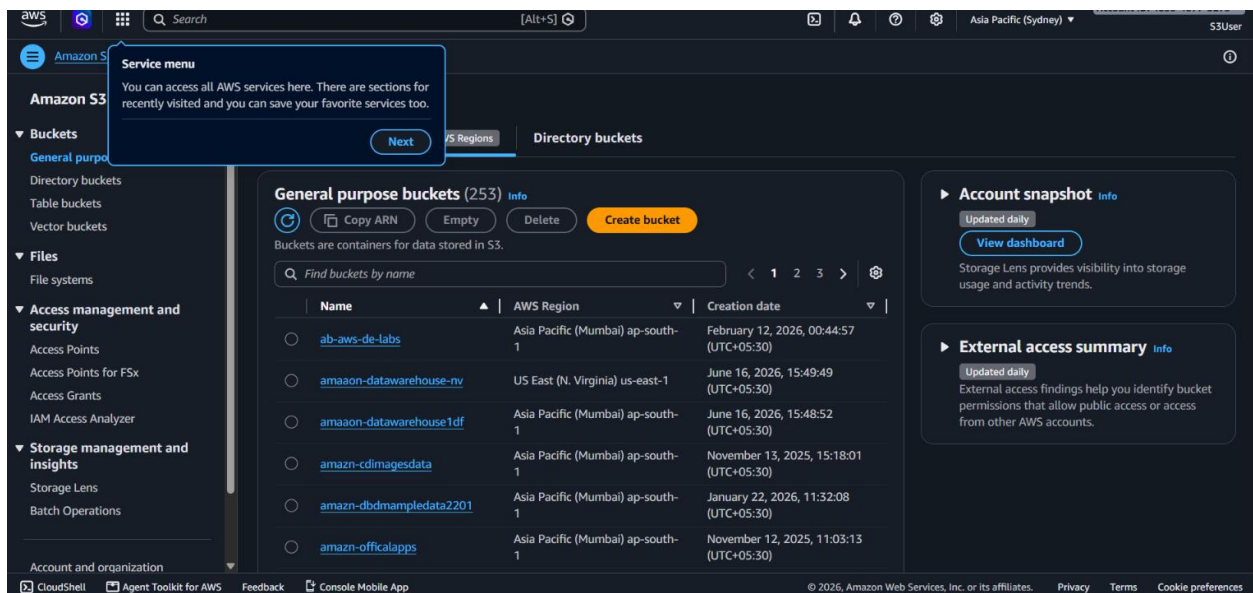
Permissions boundary (not set)

Generate policy based on CloudTrail events

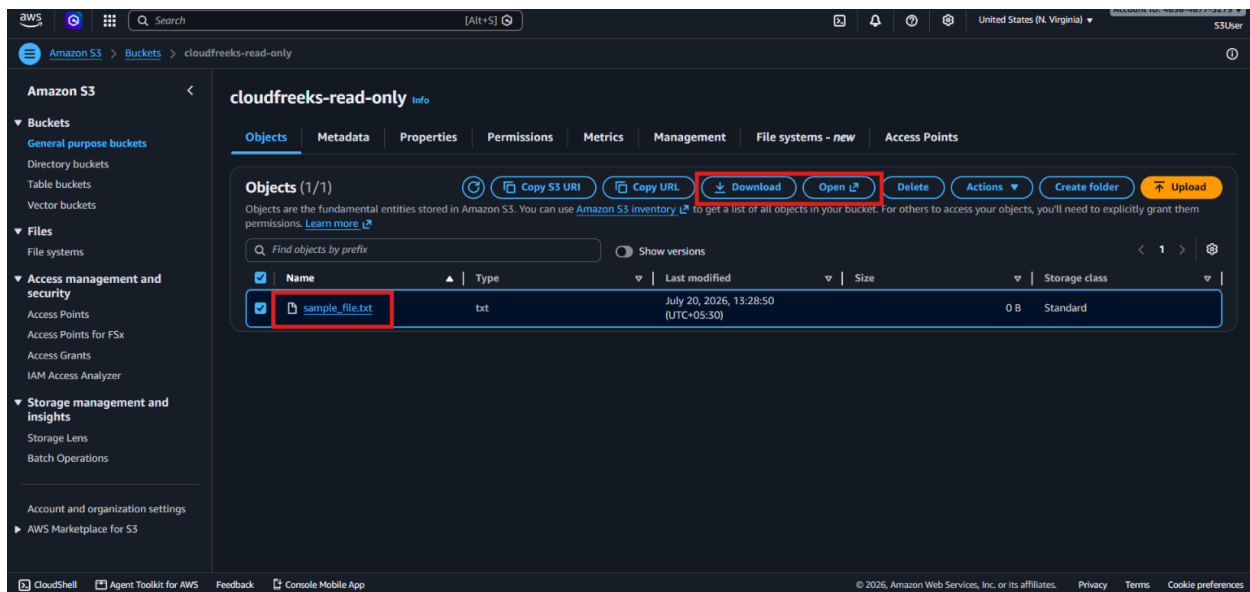
- Now attach the policy we have just created.



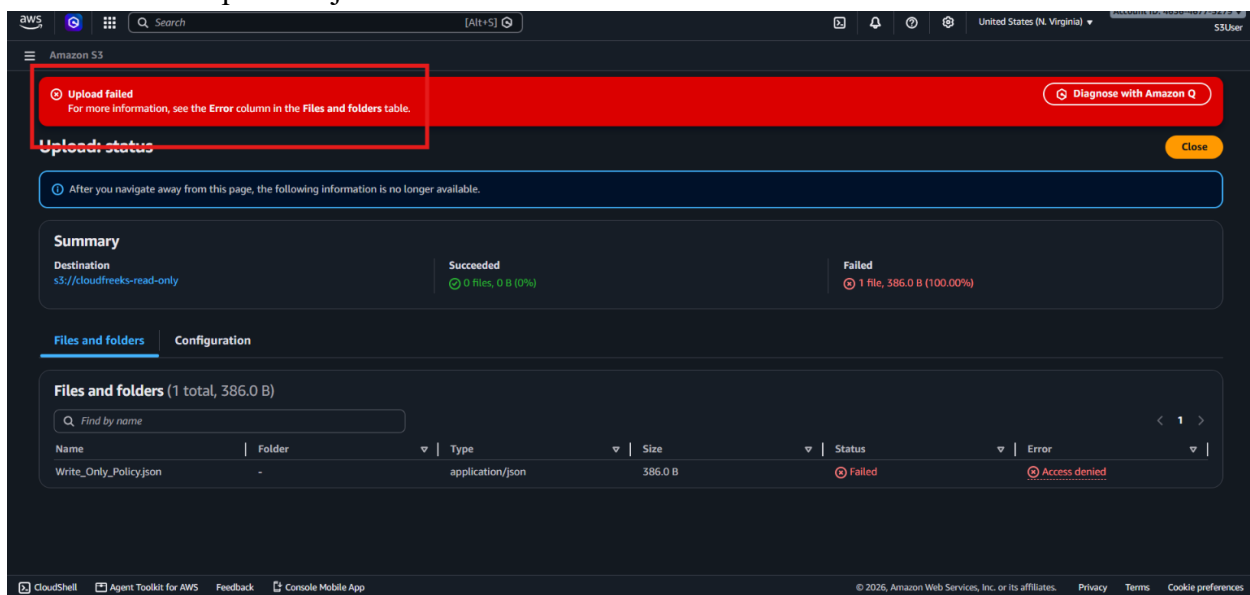
- Verify that the user can:
 - View the bucket



- Open objects
- Download/read objects



- Verify that the user **cannot**:
 - Upload objects

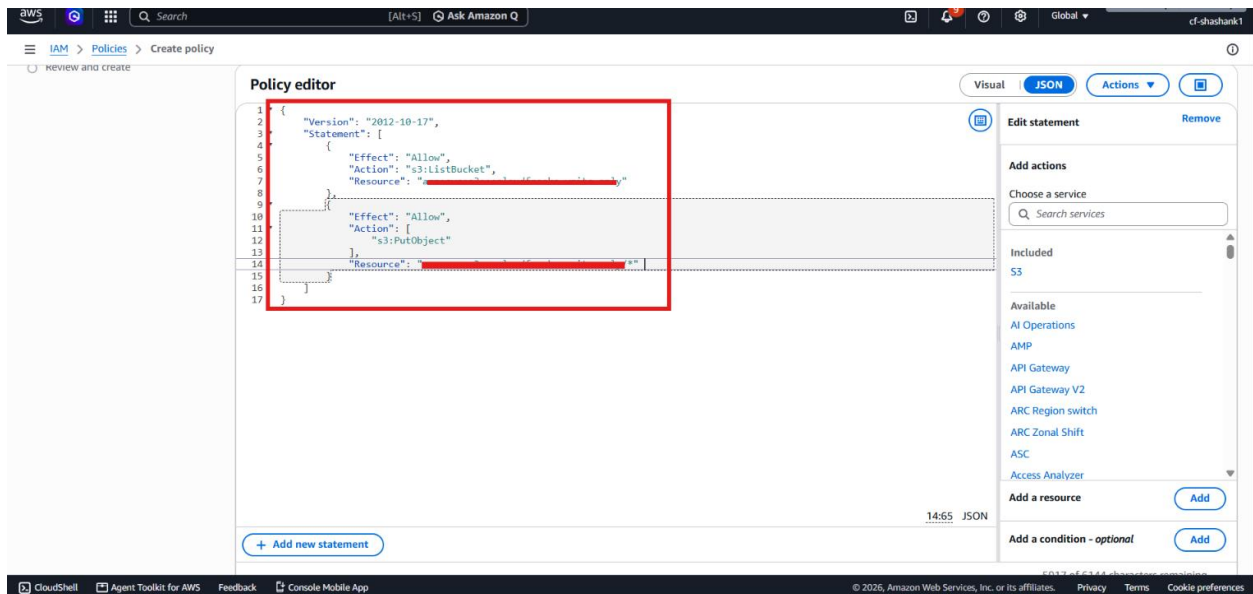


- Delete objects

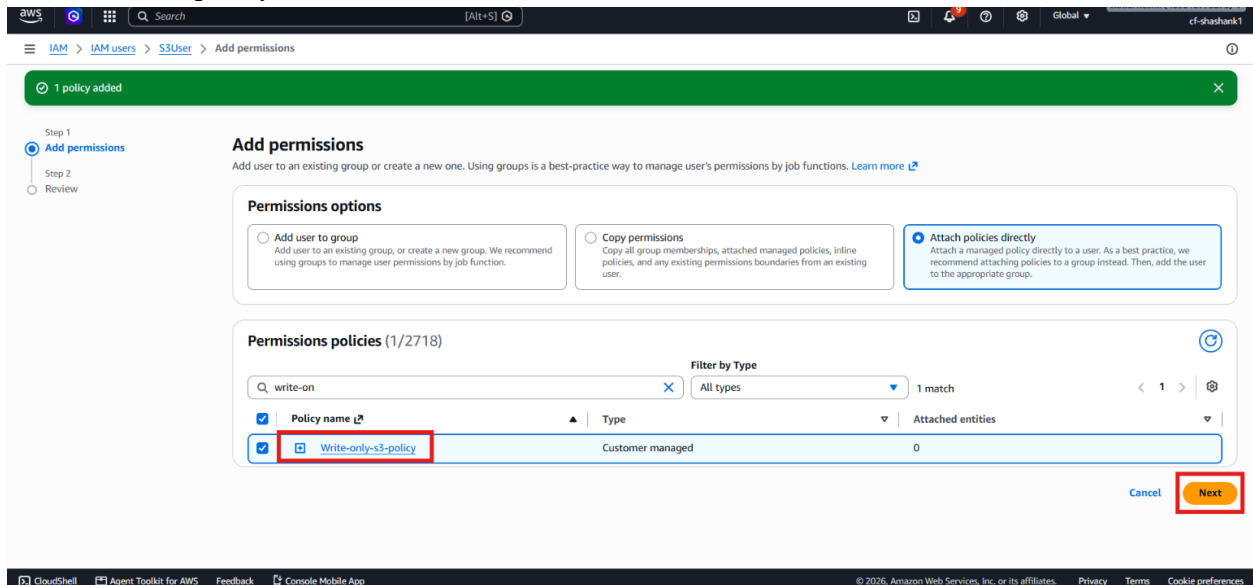
The sample JSON policy contains s3:ListAllMyBuckets, s3:GetObject, and s3:ListBucket.

Step 4: Create a Write-Only IAM Policy

- Create another Customer Managed Policy

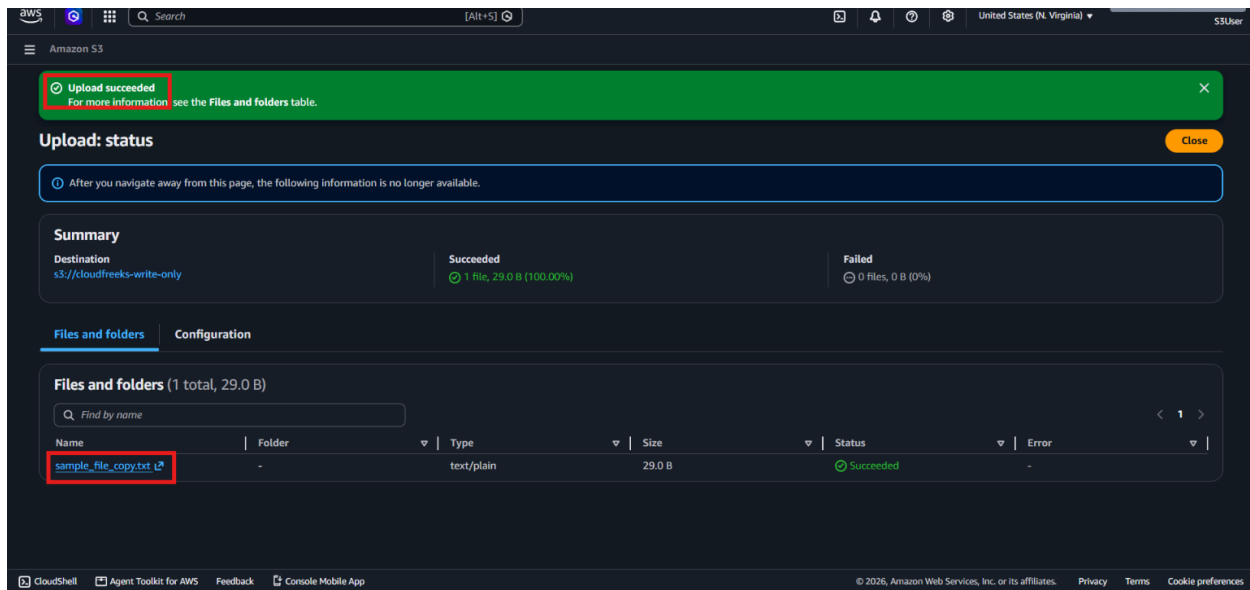


- Allow:
 - `s3:PutObject`
 - Bucket listing
- Attach the policy to the IAM user



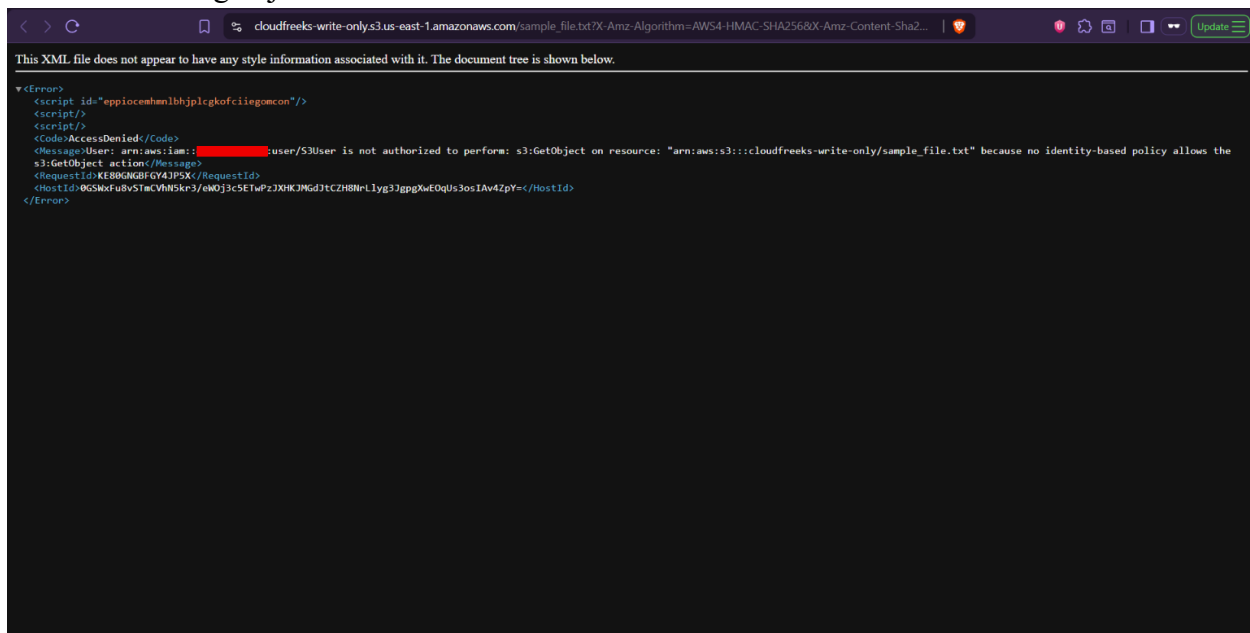
Verify that the user can:

- Upload new files



Verify that the user **cannot**:

- Read existing objects

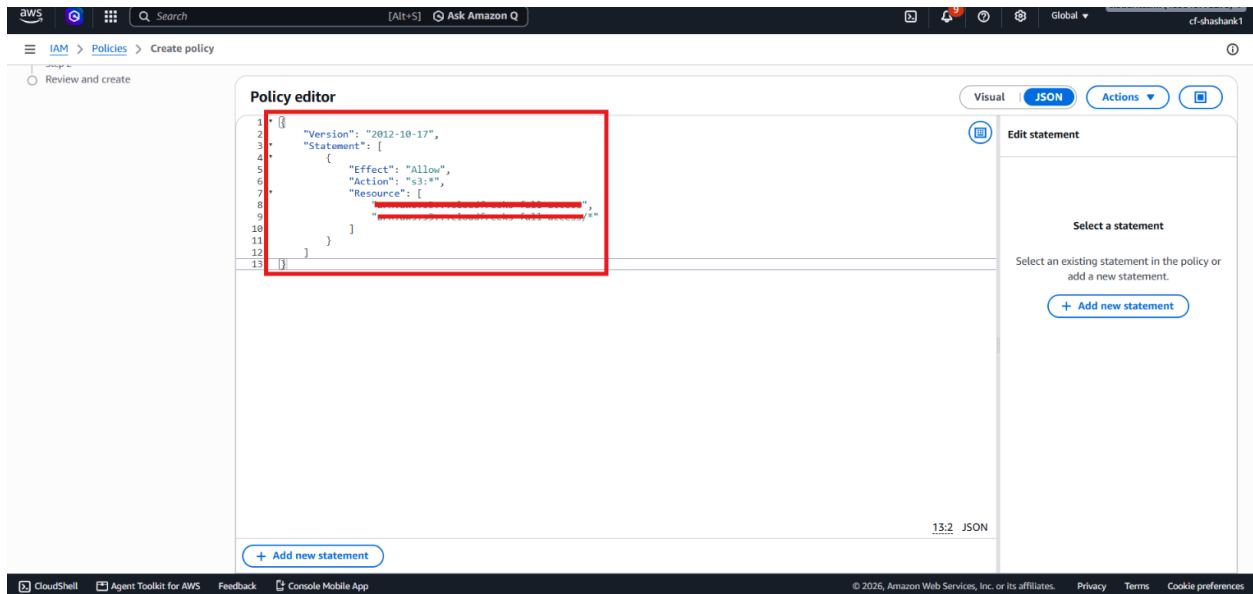


- Delete any object (including objects uploaded by the same user)

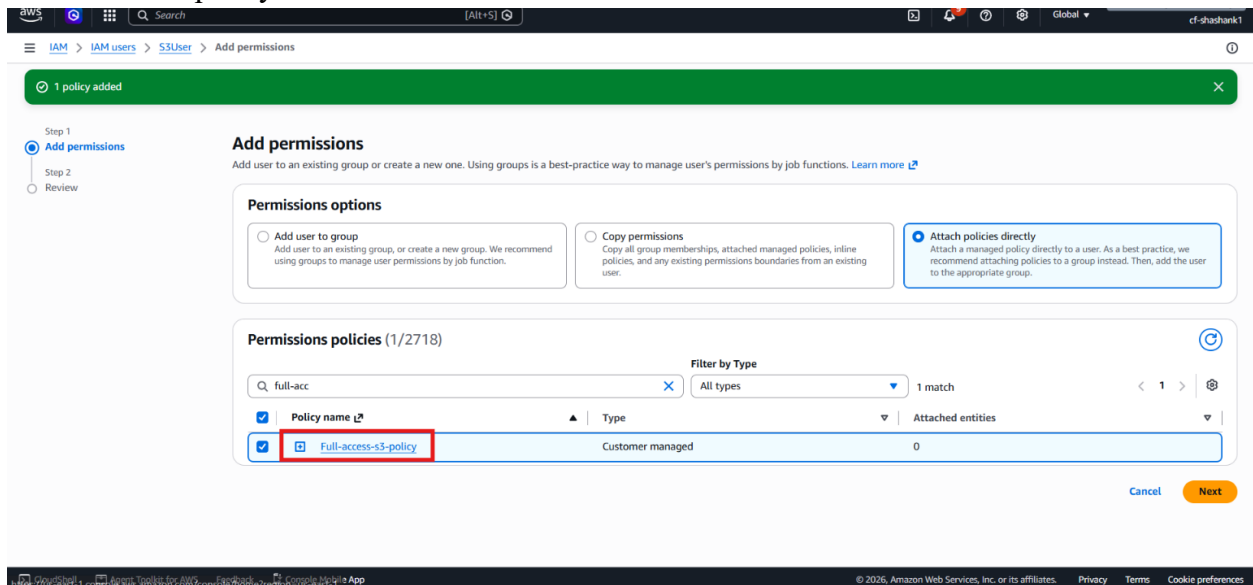
The sample JSON policy grants s3:PutObject only for the Write-Only bucket.

Step 5: Create a Full Access IAM Policy

- Create a third Customer Managed Policy

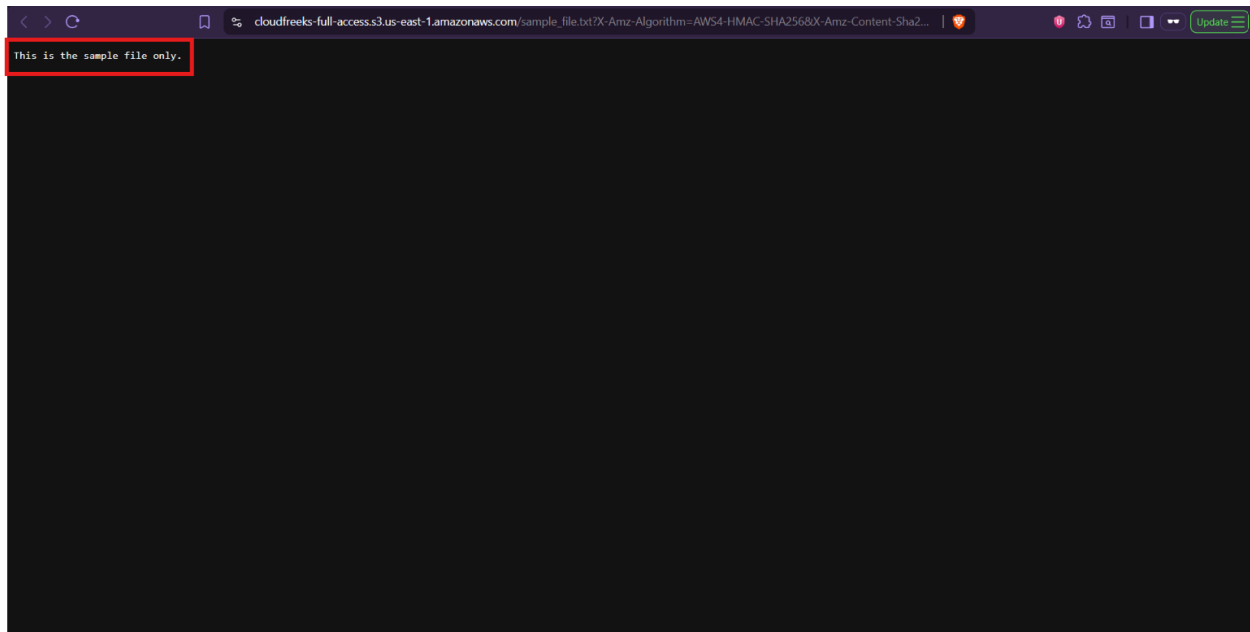


- Use the action:
 - s3:*
- Specify the ARN of the Full Access bucket
- Attach the policy to the IAM user

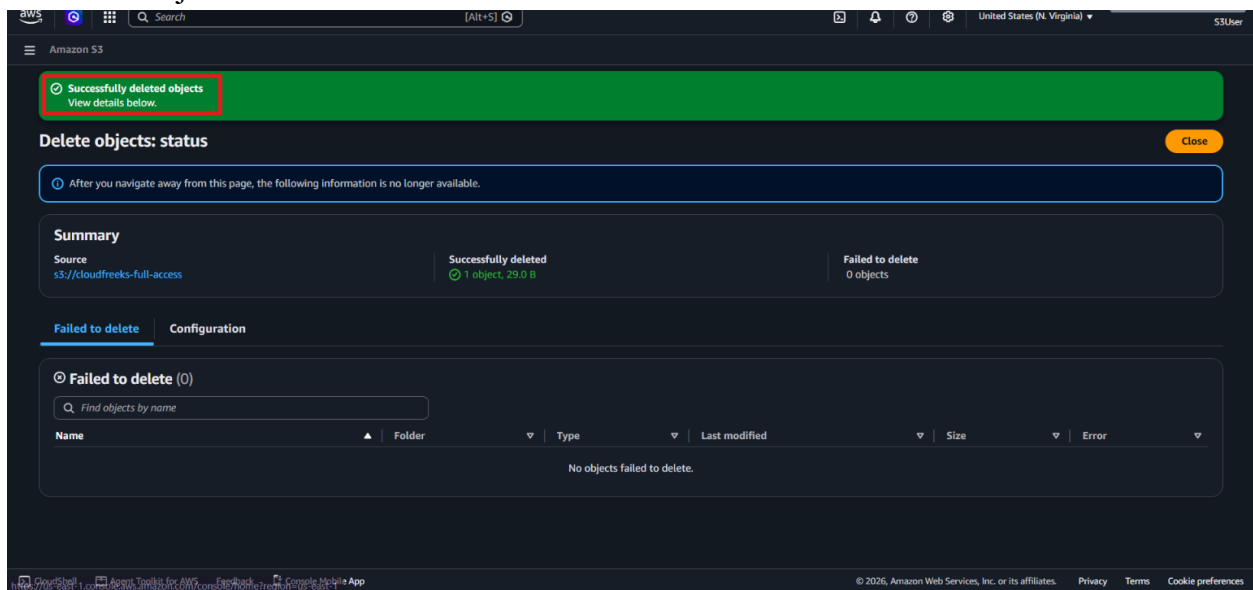


Verify that the user can:

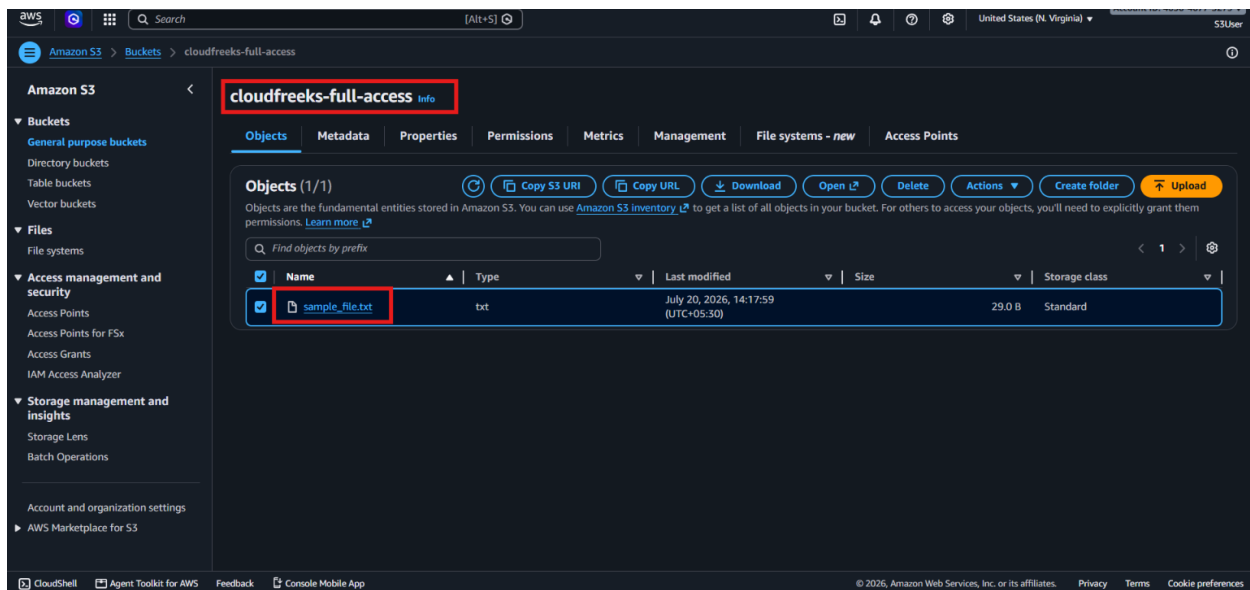
- Read objects



- Upload objects
- Delete objects



- Perform all S3 operations on that bucket

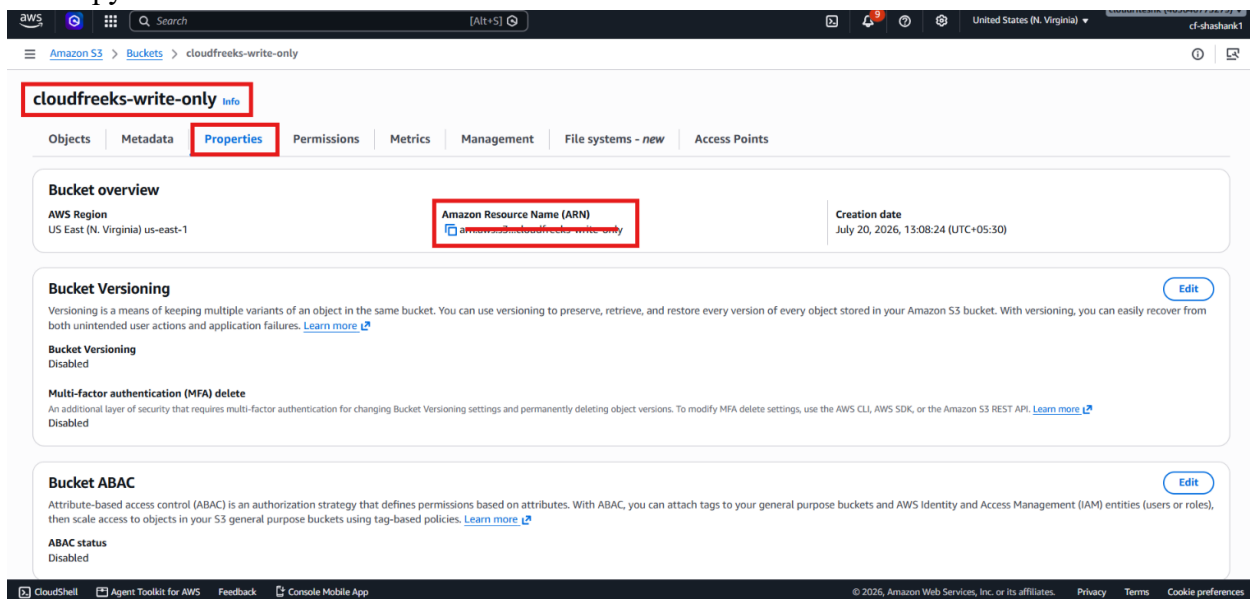


The sample JSON policy grants s3:* for the specified bucket ARN.

Step 6: Understand the Bucket ARN

For every IAM policy:

- Open the S3 bucket.
- Navigate to **Properties**.
- Copy the **Bucket ARN**.



- Replace the resource ARN inside the IAM policy JSON with the ARN of your own bucket.
- The ARN uniquely identifies the S3 bucket that the policy applies to.

Verification

- Verified that a user with no IAM policy cannot access S3 buckets.

- Verified that Read-Only policy allows object viewing but blocks uploads and deletion.
- Verified that Write-Only policy allows uploads while preventing reading and deletion.
- Verified that Full Access policy allows complete control over the specified bucket.
- Verified that IAM Policies are attached to IAM identities, while Bucket Policies are attached directly to buckets.
- Verified that the bucket ARN is required while defining S3 IAM policies.